

Holistic Detection

1. Experiment Objective

Use MediaPipe Holistic to simultaneously detect the face (468 keypoints), both hands (21 keypoints each) and body pose (33 keypoints) in the camera image.

2. Experiment Principle

1. Key Terms

Keyword	Description
MediaPipe Holistic	Google's holistic detection model, integrating three sub-models: Face Mesh (468 points) + Pose (33 points) + Hands (21 points × 2)
Three-in-one detection	Runs three MediaPipe sub-models on the same frame simultaneously, outputting full-body + hand + face keypoints together
Facial mesh	468 facial keypoints covering fine structures such as the contours of facial features, lips, eyebrows and eyelids

2. Technical Architecture

```
Terminal 1: start agent + camera + joint models
Terminal 2: start holistic detection
```

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☒ Supported
No-PTZ version	☒ Supported
Required peripherals	ESP32 camera

2. Start the agent + camera + joint models (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```

yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-10-47-55-372666-yahboom-149737
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [149778]
[INFO] [camera_driver-2]: process started with pid [149780]
[INFO] [robot_state_publisher-3]: process started with pid [149782]
[INFO] [joint_state_publisher-4]: process started with pid [149784]
[INFO] [ekf_node-5]: process started with pid [149803]
[camera_driver-2] [INFO] [1785206876.886150022] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[joint_state_publisher-4] [INFO] [1785206877.007751174] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[micro_ros_agent-1] [1785206877.173080] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785206877.754391885] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785206877.754890084] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785206877.754912957] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754918130] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785206877.754924428] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785206877.754929199] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785206877.754935071] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785206877.754939792] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785206877.754944035] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754949951] [robot_state_publisher]: got segment rfwheel_Link
[camera_driver-2] [WARN] [1785206879.871708745] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785206880.046238399] [esp32_ip_camera_publisher]: IP camera stream opened successfully

```

3. Start holistic detection (Terminal 2)

```
ros2 launch microros_v2_mediapipe_pkg holistic.launch.py
```

4. Operating Instructions

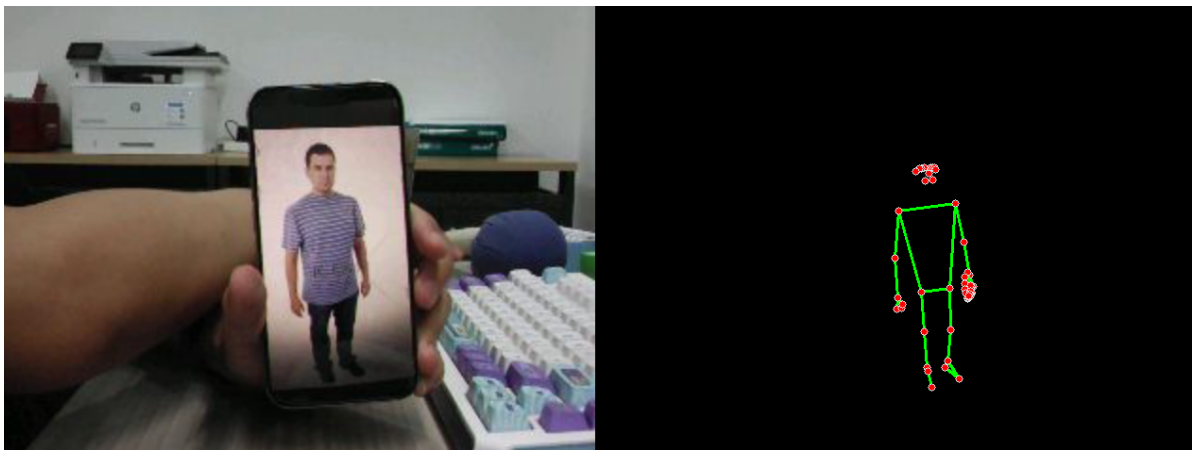
Press **q** to exit.

5. How to Stop

Press **Ctrl+C** in each terminal, one by one, to stop.

4. Observed Results

- The face is finely covered by a 468-point white mesh; each hand has 21 red dots + a skeleton; the body has a 33-point skeleton overlaid
- All three sets of keypoints render simultaneously without occluding each other



5. Code Walkthrough

Source code paths:

- Launch:


```
~/microros_ws/src/microros_v2_mediapipe_pkg/launch/holistic.launch.py
```
- Node:


```
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/Holistic.py
```

1. Core Node Analysis

```
# Source file:
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/Holistic.py
class Holistic(Node):
    def __init__(self):
        super().__init__("holistic_node")
        self.mp_holistic = mp.solutions.holistic
        self.holistic = self.mp_holistic.Holistic(
            static_image_mode=False,
            min_detection_confidence=0.5,
            min_tracking_confidence=0.5)

    def image_callback(self, msg):
        frame = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
        results = self.holistic.process(cv.cvtColor(frame, cv.COLOR_BGR2RGB))
        if results.pose_landmarks:
            self.mp_draw.draw_landmarks(frame, results.pose_landmarks,
                                         self.mp_holistic.POSE_CONNECTIONS)
        if results.face_landmarks:
            self.mp_draw.draw_landmarks(frame, results.face_landmarks,
                                         self.mp_holistic.FACEMESH_TESSELATION, ...)
        if results.left_hand_landmarks:
            self.mp_draw.draw_landmarks(frame, results.left_hand_landmarks,
                                         self.mp_holistic.HAND_CONNECTIONS)
        if results.right_hand_landmarks:
            self.mp_draw.draw_landmarks(frame, results.right_hand_landmarks,
                                         self.mp_holistic.HAND_CONNECTIONS)
        cv.imshow(self.window_name, frame)
```

Key logic:

- `mp.solutions.holistic.Holistic()`: three-in-one detector that runs the Pose, Face Mesh and Hands sub-models simultaneously
- The four landmark sets are checked independently: face (468 points), pose (33 points), left hand (21 points) and right hand (21 points) may not all be present at the same time
- `FACEMESH_TESSELATION`: triangular face mesh, finer than simple skeleton connections

2. Key Parameters

Parameter definition files:

- Launch:
`~/microros_ws/src/microros_v2_mediapipe_pkg/launch/holistic.launch.py`
- Node:
`~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/Holistic.py`

Parameter	Type	Default	Description
<code>static_image_mode</code>	bool	false	Whether to detect each frame independently
<code>min_detection_confidence</code>	float	0.5	Detection confidence threshold
<code>min_tracking_confidence</code>	float	0.5	Tracking confidence threshold

3. Node Description

Node	Function
<code>holistic_node</code>	MediaPipe Holistic three-in-one face + hand + pose detection

6. Frequently Asked Questions

Issue	Cause	Solution
Low performance / lag	Holistic runs three models simultaneously, which is computationally heavy	Detect only the upper body (don't raise your hands) to reduce the cost of face + hands appearing together
Only part of the keypoints show	The corresponding body part is not in the frame	Make sure the whole body + face + hands are all within the camera's field of view